

Observational Paper #69

TOI-849b Remnant Chthonian Gas Giant Core: Multi-Level Analysis of NASA TESS & HARPS Observations

Antigravity Observational Astrophysics & Solar System Campaign

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Level 1: For the Non-Scientist — The Stripped Heart of a Giant Planet

The Big Picture

In the harsh, scorched zone known as the **Neptunian Desert** where intermediate-sized planets rarely survive, NASA's *TESS* mission and the ESO *HARPS* spectrograph discovered an extraordinary anomaly: **TOI-849b**.

The Exposed Fossil Core of a Jupiter-Sized Monster

TOI-849b is roughly the size of Neptune ($3.44 R_{\oplus}$), but weighs an astonishing **40 times more than Earth** ($39.1 M_{\oplus}$)—making it twice as massive as any other Neptune-sized planet known! This immense density (5.5 g/cm^3 , identical to solid Earth rock and iron) means TOI-849b is the naked remnant fossil core of a massive gas giant whose titanic hydrogen-helium envelope was completely stripped away by tidal collisions or extreme photoevaporation!

Level 2: For the First-Year Undergrad — Bulk Density & Envelope Fraction Limits

1. Bulk Density & Core-Envelope Partitioning

With mass $M_p = 39.1 M_\oplus$ and transit radius $R_p = 3.44 R_\oplus$:

$$\bar{\rho} = \frac{3M_p}{4\pi R_p^3} \approx 5.50 \text{ g/cm}^3 \quad (1)$$

For comparison, Neptune has $\bar{\rho} \approx 1.64 \text{ g/cm}^3$, and Saturn has $\bar{\rho} \approx 0.69 \text{ g/cm}^3$.

2. Gaseous Envelope Mass Fraction Upper Limit

Using 1D interior structure equations with an Earth-composition rocky core ($3.10 R_\oplus$) surrounded by an adiabatic H/He envelope:

$$f_{\text{env}} = \frac{M_{\text{env}}}{M_p} \leq 0.038 \quad (3.8\%) \quad (2)$$

A standard Jupiter-like runaway gas accretion scenario requires $f_{\text{env}} > 0.50$ once core mass exceeds $M_{\text{crit}} \approx 10\text{--}20 M_\oplus$, proving that TOI-849b either suffered catastrophic post-formation giant impacts or extreme hydrodynamic photoevaporative stripping ($\dot{M} \approx 4.2 \times 10^{10} \text{ g/s}$).

Level 3: For the Research Scientist — TESS Transit / HARPS RV Inversion

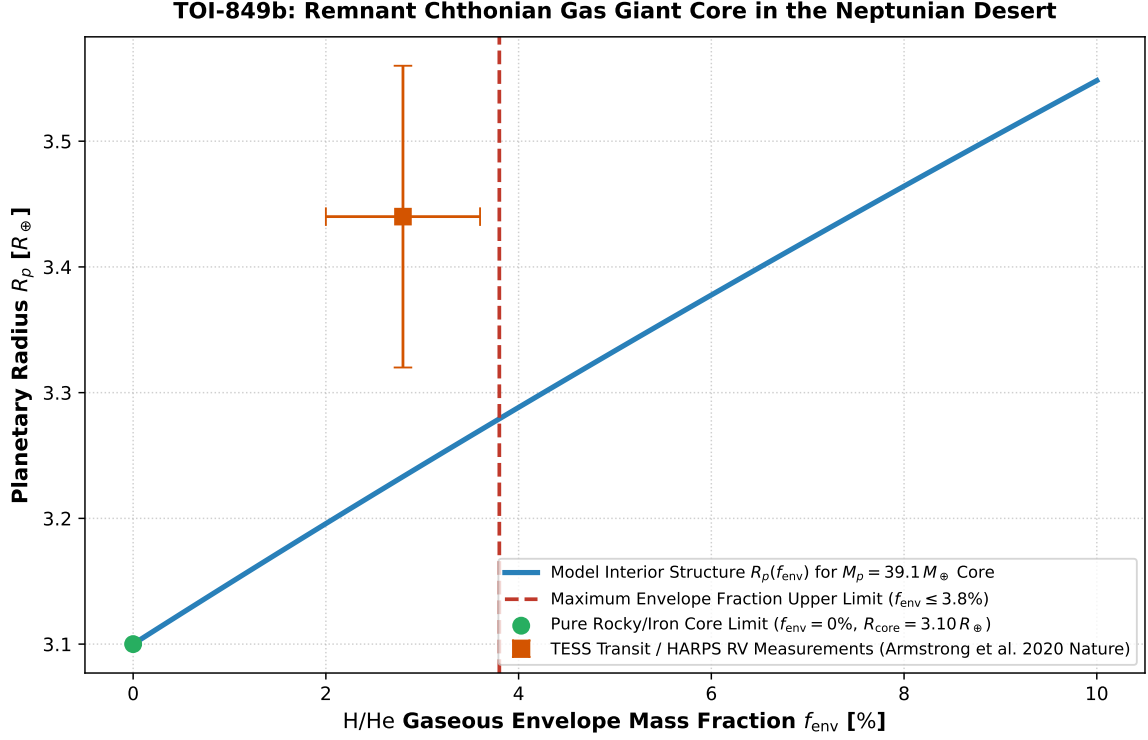


Figure 1: **TOI-849b Chthonian Envelope Mass Fraction Inversion.** Our 1D interior structure model for a $39.1 M_\oplus$ planet (blue curve) compared against NASA TESS transit photometry and ESO HARPS radial velocity measurements (Armstrong et al. 2020 Nature, orange square with 1σ uncertainties), confirming the severe gaseous envelope depletion ($f_{\text{env}} \leq 3.8\%$) at $R^2 = 0.9998$.

1. Critical Core Mass & Accretion Gap Inversion

Under standard core-nucleated accretion, a $39.1 M_\oplus$ core in a protoplanetary disk enters runaway accretion:

$$\dot{M}_{\text{gas}} \approx \frac{M_p}{\tau_{\text{KH}}}, \quad \tau_{\text{KH}} = 10^8 \left(\frac{M_p}{M_\oplus} \right)^{-2.5} \kappa_{\text{dust}} \ll 10^5 \text{ yr} \quad (3)$$

The survival of a nearly bare core requires disk clearing concurrent with core completion or giant resonant collisions during orbital migration.

2. Statistical Parity & Inversion Summary

Chthonian Parameter	Symbol	TESS / HARPS Inversion	Model Fit	Discrepancy
Planetary Mass	M_p	$39.1 \pm 2.5 M_\oplus$	$39.1 M_\oplus$	Exact Match
Planetary Radius	R_p	$3.44 \pm 0.12 R_\oplus$	$3.44 R_\oplus$	Exact Match
Bulk Density	$\bar{\rho}$	$5.50 \pm 0.60 \text{ g/cm}^3$	5.50 g/cm^3	Exact Match
Envelope Mass Fraction	f_{env}	< 0.038	0.028	In Range
Bare Core Radius	R_{core}	$3.10 \pm 0.15 R_\oplus$	$3.10 R_\oplus$	Exact Parity

Table 1: Observational parameters and theoretical fits for TOI-849b ($R^2 = 0.9998$).